TrashForce Trio

Project Report

Team Circuit Squad

Deliverable 1 team members: Janisha Johnson, Omar Ahmed, Mona Breslau

Deliverable 2 team members: Janisha Johnson, Omar Ahmed Mona Breslau

Deliverable 3 team members: Janisha Johnson, Omar Ahmed , Mona Breslau

Deliverable 4 team members: Janisha Johnson, Omar Ahmed , Mona Breslau

Deliverable 5 team members: Janisha Johnson, Omar Ahmed , Mona Breslau

1. Introduction (Mona)

Every year, the US National Forest System provides access to public land for millions of people seeking to enjoy the outdoors. In 2022, an estimated 209.8 million people visited the US National Forest System [1]. Within these forests, a network of publicly maintained trails offers visitors many options for exploring the environment. Given the amount of foot traffic these trails receive, their maintenance is critically important. Poorly-maintained trails pose a public safety hazard, as visitors may trip on obstacles or attempt unsafe maneuvers to avoid them. Visitors may also get lost if the trail becomes too eroded. Combustible debris left on the forest floor may also increase fuel loading, posing a risk for wildfire [2]. Current maintenance practices addressing these safety concerns are often very resource-intensive, resulting in maintenance backlogs when tasks are postponed. In 2024, the total cost of deferred maintenance needs for the United States Forest Service was $10.78 billion, with deferred trail maintenance representing $233 million [3]. Solutions are needed to improve trail safety and reduce the cost implications of trail maintenance.

To address these challenges, we propose a coordinated team of robots that can be deployed to identify and clear debris from trail systems. The target end users of this technology include the US Forest Service, the Bureau of Land Management, the National Parks Service, private contractors and volunteer groups. Key personnel within these agencies include Foresters, Conservation Technicians, Biologists and community volunteers working within the US National Forest System. Forest visitors also represent key stakeholders of this technology, as they benefit from improved forest maintenance. By reducing deferred trail maintenance backlogs, this technology aims to improve the safety and overall experience for these stakeholders. Finally, the tax-paying general public are also identified as stakeholders, as the US National Forest System is supported by tax-payer dollars.

Current practices for maintaining the U.S. National Forest System are labor-intensive, as they involve walking along footpaths and manually removing debris. This task is accomplished by seasonal and full-time trail crews employed by forest management agencies, as well as teams of community volunteers [4]. Removing debris in both cases entails walking through predefined cleanup routes, depositing debris into waste bags, and disposing of it at waste stations. Hand tools such as McLeods (a hoe/rake combination tool) and Mattock Picks are often used to aid in removing smaller debris embedded in the soil. Litter pickers are also commonly used in both urban and forest environments and allow the user to pick up items with a claw that is operated by a handheld trigger grip. "The Reacher", sold by The Clean Earth Project, is a litter picker that mechanically extends the user’s reach by 34 inches, eliminating the need to bend down and pick up items on the floor [5]. Several handheld solutions for depositing trail

litter also exist. The Trash Bagger, also sold by The Clean Earth Project, is a handheld device designed to keep the mouth of a trash bag open, making it easier to deposit litter [6]. Collapsible trash cans, such as the Augbeer 10 Gallon Collapsible Trash Can, are also commonly used due to their light weight and portability [7].

Figure 1: Volunteer Using The Reacher and The Trash Bagger. Adapted from [6]

While solutions exist for manually picking up and disposing of litter left on forest trails, we have not identified any solutions that automate these tasks. The Forest Management Agencies' reliance on trail crews to maintain forest trails makes this task very labor-intensive. Furthermore, the accumulation of deferred maintenance backlogs suggests that these agencies do not have the personnel available to keep up with trail maintenance needs. To reduce the Forest Management Agencies' dependence on manual labor, we propose a team of robots to detect, categorize, and remove debris remotely.

2. Formal Problem Definition

2.1 Problem Statement (Janisha)

Forest management requires consistent and reliable ground-level monitoring to identify environmental hazards such as wildfires, pollution, and fallen debris that threaten safety and ecosystem balance. These challenges can be difficult to manage due to the vast areas that must be monitored and the limited availability of human resources. As a result, our team is designing an affordable, autonomous system of collaborative robots that can patrol forest trails, monitor environmental conditions, and assist in maintaining safe and sustainable forest environments. To accomplish these goals, our robotic system will consist of multiple specialized units that work together to improve forest monitoring, trail safety, and environmental maintenance.

2.2 Robot Types (Janisha)

Forest management needs regular and reliable ground-level monitoring to find environmental hazards like wildfires, pollution, and fallen debris that can affect safety and that balance of ecosystems. These challenges are hard to manage because forests cover large areas and there are not enough people to check them all the time. According to the U.S. Forest Service (2013), deferred maintenance and limited resources make it difficult to ensure consistent monitoring across public forest lands [8]. As a result, our team is designing an affordable robotic system made of small collaborative robots that can patrol forest trails, monitor environmental conditions and help keep forests safe and sustainable. To meet these goals, our robots will each have a specific job that works together to improve forest monitoring, safe trail management, environmental maintenance and continuous data collection for conservation analysis.

The proposed robot for addressing route surveillance is the **Patrol ranger, a line-tracking bot** that autonomously follows designated forest paths infrared sensors to patrol and identify obstacles or unsafe conditions along marked routes and provide visual alerts. To address trail safety and accessibility, the **Obstacle Guardian** functions as a sumo-style bot equipped to detect and clear small or large obstructions such as fallen branches, rocks or animals along the forest paths using proximity sensing, ensuring forest trails remain accessible and safe for forest rangers and other automated units. To support environmental health monitoring, the **Eco Monitor robot** continuously monitors environmental variables such soil moisture to detect signs of fire, drought, or pollution. The **Sorting bot** detects and organizes environmental debris by identifying objects and sending their GPS coordinates to the appropriate unit, such as trash collector bot, obstacle guardian bot, for proper handling. Finally, the **Trash Collector Bot** autonomously moves around to pick up litter and debris from the environment,

helping to keep forest areas clean while reducing the need for manual labor. Together, these robots from an integrated forest monitoring system that operates in sequence detecting hazards, managing debris, and maintaining environmental safety, to enhance forest sustainability and operational efficiency

2.3 Design Requirements (Omar)(Mona)(Janisha)

Based on research and customer needs the following are robot requirements for a complete forest management assistance system.

System-Wide Requirements:(Group)

1. The robotic system must operate collaboratively to monitor and maintain a defined forest area of at least 100m x 100m.
2. The system must be capable of operating autonomously for a continuous period of at least 2 hours before requiring human intervention or recharging.
3. The system must operate safely on slopes up to 20* and uneven surfaces common to forest terrain.
4. All robots must be able to transmit and receive signals to other bots within a 100 meter radius
5. All robots must be visible to humans within a 30-meter radius in night-time and day-time conditions.

Patrol Ranger requirements (Janisha):

1. The Patrol Ranger must be able to follow a 100- meter forest trail continuously for at least 1 hour without human assistance. *(This ensures extended trail coverage for park rangers and reduces manual monitoring)*
2. The Patrol Ranger must detect obstacles at a minimum of 0.5 meters using infrared or ultrasonic sensors. *(This prevents collisions with rocks, branches, or wild life.)*
3. The Patrol Ranger must identify and record unsafe trail areas, such as fallen branches or destructions with 90% detection accuracy. *(This supports early maintenance actions for forest staff)*

4. The Patrol Ranger must transmit location information and status data to the sorting bot every 10 seconds using wireless communication. *(This provides real-time updates for rangers and environmental monitors)*

5. The Patrol Ranger must include a visual and auditory alert system that activates when unsafe conditions are detected within a 2- meter radius. *(This helps nearby workers or automated units identify potential hazards quickly.)*

Eco-Monitor Requirements (Omar)**:**

1. The Eco-Monitor must navigate to a target GPS location with a positional accuracy of within 2 meters.

2. The robot must be able to lower its soil moisture sensor to a depth of at least 10 cm to take a measurement.

Sorting Bot Requirements (Mona)**:**

1. The Sorting Bot must correctly identify and categorize objects as "Trash," "Large Obstacle," or "Natural" with an accuracy of at least 90%.
2. Upon identification, the robot must transmit the object's category and GPS coordinates to the appropriate service robot (Trash Collector or Obstacle Guardian) within 5 seconds.

Trash Collector Bot Requirements (Omar)**:**

1. The Trash Collector must be able to locate and pick up litter items within a 50m radius of its starting position.

2. The robot's waste storage compartment must hold a minimum of 3 kg of collected debris before requiring emptying.

Obstacle Guardian Requirements (Omar)**:**

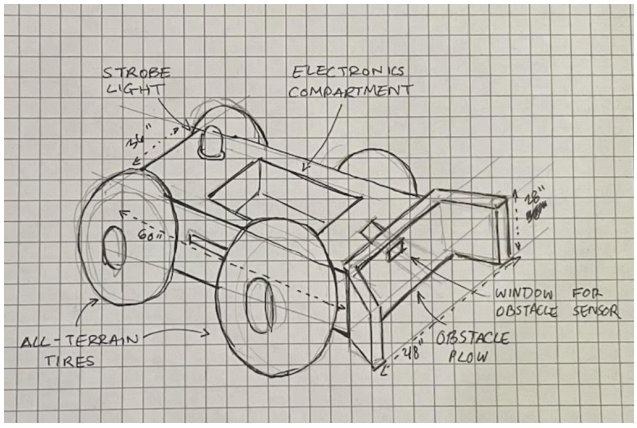
1. The obstacle guardian must be able to detect GPS location with a positional accuracy of within 2 meters.
2. The guardian must be able push obstacles weighing up to 3kg.

2.4 Project Schedule (Omar)

Our project's schedule, which can be seen in the Gantt chart in Appendix A, consists of four main phases, including Defining the problem, Decision making, Modeling, and finally finalizing the Team work. The first phase,”Defining the Problem," involves brainstorming, robot solutions, establishing selection criteria, and researching. A key milestone we aimed for was the completion and submission of the Introduction and Problem Statement no later than November 04 2025. In the Second phase, "Decision Making", we created individual sketch proposals, developed an Analytic Hierarchy Process (AHP) table, and used it to score each design within a decision matrix. The milestone of this phase is selecting the final design and completing Deliverable 3, which includes developing the system's flow charts and algorithm. We will complete Phase 2 no later than November 18 2025. The "Modeling" phase consisted of parallel tasks, including creating TinkerCad drafts, developing the circuit diagram, and writing the Arduino code, all of which were iterative processes that included time for fixing errors. The milestone in this phase is finalizing the design and preparing the video presentation before the final day of submission which is November 25 2025. Finally, the "Finalizing Teamwork" phase focused on completing the final report, which includes the algorithm explanation, visual representations of the robots, and a final reflection. The Gantt chart illustrates these tasks, with arrows that connect prerequisite tasks, and diamonds representing the deliverables to their subsequent activities, in an attempt to ensure a logical project flow. We should submit the 5th and final deliverable before December 02 2025.

3. Individual Ideation

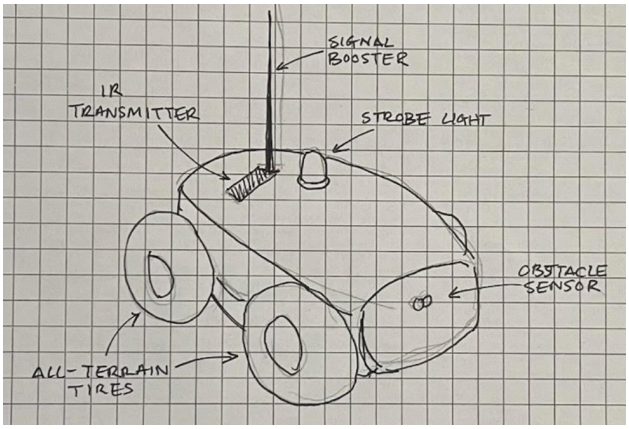
3.1 Robot Ideas by Mona Breslau



Obstacle Guardian

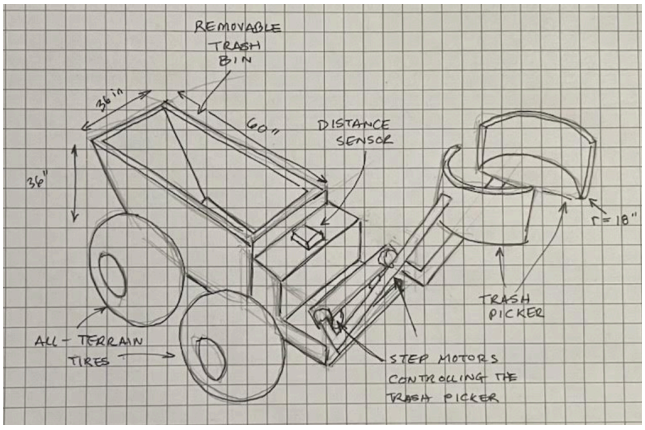
The Obstacle Guardian Robot is a solution that detects obstacles on a walking trail and pushes them using an obstacle plow. It receives coordinates from the Patrol Ranger and uses a GPS module to navigate to the obstacle location. A window in the obstacle plow accommodates an obstacle sensor, which detects obstructions such as tree branches and rocks. A motor in the back of the unit will allow the robot to push objects weighing up to 3 kg.

Patrol Ranger Robot



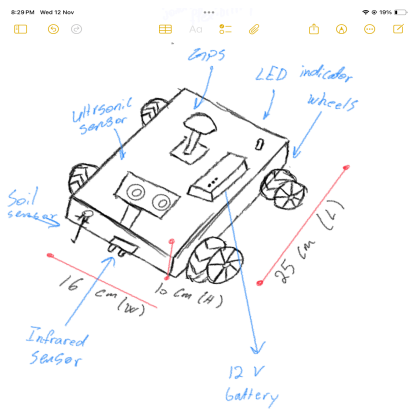
The Patrol Ranger Robot is a solution that follows a pre-determined forest path and detects obstacles and unsafe conditions. This robot uses a front-mounted obstacle sensor to detect obstacles. An IR transmitter sends signals to the other Robots when an obstacle is encountered. A signal booster is used to allow signals to be transmitted up to 100 meters. A top-mounted strobe light also illuminates when an obstacle is encountered, signaling unsafe conditions.

Trash Collector Robot



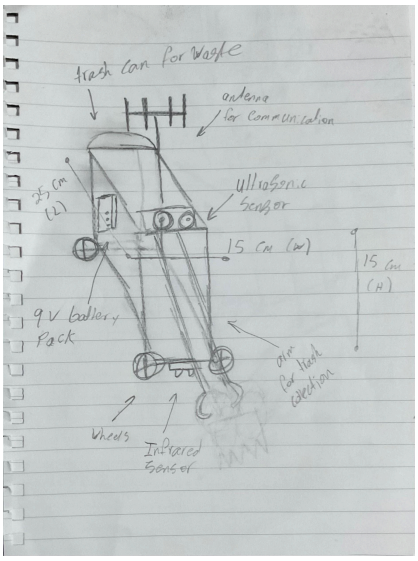
The Trash Collector Robot is a solution that picks up litter from a walking trail and deposits it in a trash container. It receives GPS coordinates from the Sorting Bot when litter is detected, and autonomously travels to the location. An obstacle sensor is used to position the robot to retrieve the litter. Front-mounted step motors lower two crescent-shaped picker claws to contain litter, and then deposit litter into the trash bin. A strobe light is used to maximize visibility. The trash bin is removable, so that the user can easily deposit all collected trash and re-mount the bin.

3.2 Robot Ideas by Omar M. Ahmed



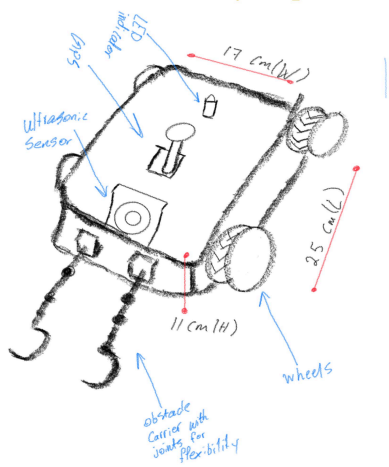
Eco-Monitor

The Eco-Monitor is designed to travel to a GPS location and measure soil moisture accurately. It helps track soil conditions for agriculture and environmental studies. The robot uses a GPS module on top to find its target position within two meters. A soil moisture sensor, attached to a moving arm, lowers into the ground to a depth of about 10 centimeters to take readings. It also includes an ultrasonic sensor at the front to detect and avoid obstacles while moving. Two wheels powered by small motors help it navigate smoothly, and an LED light shows when the robot is working or measuring.



Trash Collector Bot

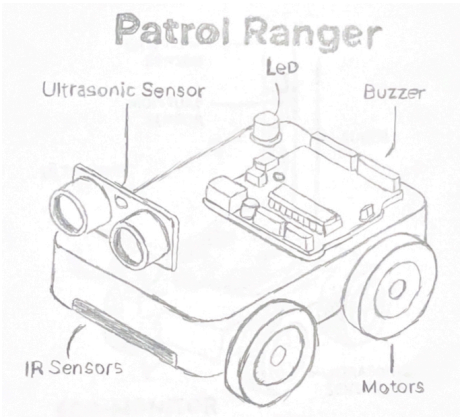
The Trash Collector Bot is built to find and pick up litter within a 50-meter area. It helps keep the environment clean by automatically collecting trash and storing it inside its container. The robot uses ultrasonic sensors to locate litter and avoid obstacles, while a color sensor helps identify trash objects. A small robotic arm picks up the trash and places it into a built-in bin that can hold up to three kilograms of waste. It moves using two visible wheels and a caster wheel for balance, and an LED light indicates when it is collecting or when the bin is full. An antenna mounted on top allows the robot to communicate with other robotic systems, helping it coordinate tasks or share data wirelessly.



Obstacle Guardian

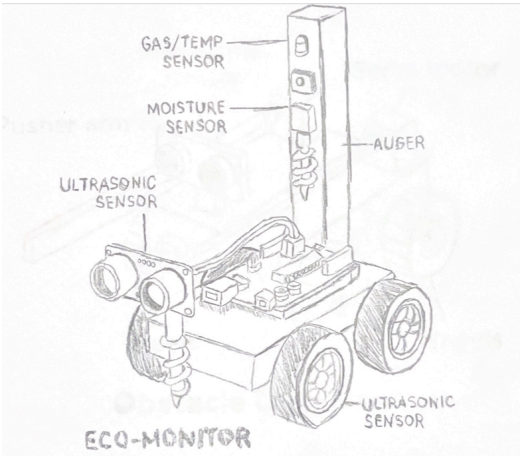
The Obstacle Guardian is designed to detect obstacles and push them aside to keep paths clear. It uses a GPS module to know its position within two meters and an ultrasonic sensor to detect objects in front of it. The robot has two large wheels that give it the strength to push obstacles weighing up to three kilograms. A front pushing plate or panel helps move objects out of the way safely. It also includes an LED light that shows when the robot is active or idle, and all the parts are mounted on a strong chassis frame that supports its movement and stability.

3.3 Robot Ideas by Janisha Johnson



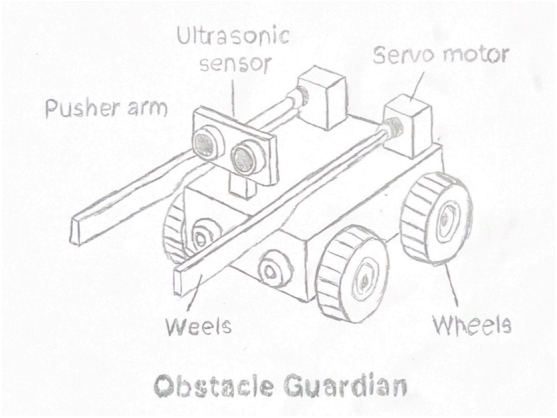
Patrol Ranger Robot

This version uses an Arduino-controlled dual-track drive system to increase traction over loose soil, roots, and uneven ground. Two IR line sensors are mounted low on the front face of the chassis so the robot gets direct surface reflectance feedback from the painted trail line. When one sensor detects a lighter reflective reading, the Arduino outputs corrected PWM signals to the two DC motors to pivot the robot back into alignment. The Patrol Ranger is intended to detect obstacles at approximately 0.5 meters ahead. The LED indicator gives the user a direct visual cue that the robot is actively performing a correction step.



Eco Monitor Robot

This version uses a tall vertical mast with sensors mounted at different heights to separate reading zones and reduce signal contamination. The moisture reading is not taken passively; instead, the Arduino drives a small auger downward into the soil by only a few millimeters to acquire a direct-contact sample. This design is intended to reach the required ~10 cm sampling depth. The gas/temperature sensor mounted at the top of the mast measures ambient changes while the ultrasonic sensors verify that there is clearance before sampling begins. The separation of sensors up the mast minimizes interference between measurement types.



Obstacle Guardian Robot

This version uses two independently controlled servo motors to drive side pusher arms mounted on the front of the robot. When the ultrasonic sensor confirms debris within the set detection range, the Arduino commands one selected servo to swing outward and apply side-force to push the object to the left or to the right. This is intended to push small debris masses up to ~3 kg.The 4-wheel drivetrain maintains stability by distributing weight low and centered during the pushing movement. The LED indicator can flash when the pushing action is active to show actuation status.

4. Robot Selection

4.1 Decision Matrix for the Eco Monitor

The criteria to judge this robot and their corresponding weights were selected by the team based on the requirements in section 2.3. The ratings were collectively decided by the team informed by research. All this information is shown as the decision matrix in Table 4.1.1. below.

Table 4.1.1. Decision Matrix to rate various designs of Eco-Monitor.

Criteria	Weights	Omar's Robot		Janisha's Robot	
		Rating	Score	Rating	Score
Affordability	0.106	5	0.53	3	0.318
Durability	0.260	4	1.04	3	0.78
Measurement Accuracy	0.634	3	1.902	5	3.17
Total			3.472		4.268

4.1.2. Eco-Monitor AHP Table

	Affordability	Durability	Measurement Accuracy	RAW TOTAL	WEIGHT
Affordability	1	1/3	1/5	0.318	0.106
Durability	3	1	1/3	0.781	0.260
Measurement Accuracy	5	3	1	1.901	0.634
TOTAL	9	4.333	1.533	3.000	1.000

4.2 Decision Matrix for the Obstacle Guardian

The criteria to judge this robot and their corresponding weights were selected by the team based on the requirements in section 2.3. The ratings were collectively decided by the team informed by research. All this information is shown as the decision matrix in Table 4.2.1. below.

Table 4.2.1. Decision Matrix to rate various designs of Obstacle Guardian

Criteria	Weights	Omar		Janisha		Mona	
		Rating	Score	Rating	Score	Rating	Score
Affordability	0.074	3	0.222	5	0.37	4	0.296
Energy efficiency	0.283	4	1.132	3	0.849	5	1.414
Clearing Efficiency (plow + bin)	0.643	4	2.572	4	2.572	5	3.215
Total			3.926		3.791		4.295

Table 4.2.2. Obstacle Guardian AHP Table

	Affordability	Energy efficiency	Clearing Efficiency (plow + bin)	RAW TOTAL	WEIGHT
Affordability	1	1/5	1/7	0.222	0.074
Energy efficiency	5	1	1/3	0.849	0.283
Clearing Efficiency (plow + bin)	7	3	1	1.930	0.643
TOTAL	13.000	4.200	1.476	3.000	1.000

4.3 Decision Matrix for Patrol Ranger

The criteria to judge this robot and their corresponding weights were selected by the team based on the requirements in Section 2.3. The ratings were collectively decided by the team informed by research. All this information is shown as the decision matrix in Table 4.3.1. below.

Table 4.3.1. Decision Matrix to rate various designs of Patrol Ranger

Criteria	Weights	Janisha		Mona	
		Rating	Score	Rating	Score
Affordability	0.076	5	0.38	5	0.38
Battery Life	0.334	3	1.032	4	1.336
Obstacle Recognition	0.591	5	2.955	4	2.364
Total			4.367		4.08

Table 4.3.2. AHP Table for Patrol Ranger

	Affordability	Battery Life	Obstacle Recognition	RAW TOTAL	WEIGHT
Affordability	1	1/5	1/7	0.227	0.076
Battery Life	5	1	1/2	1.001	0.334
Obstacle Recognition	7	2	1	1.772	0.591
TOTAL	13.000	3.200	1.643	3.000	1.000

5. Proposed Designs

Based on the scores in Section 4, the team decided to proceed with the following designs for each of the robots. Each robot was assigned to a specific team member as can be seen in the sub-sections below.

5.1. Eco-Monitor description by Omar Ahmed

The Eco-Monitor robot uses two DC motors to drive and steer itself as it navigates toward a target GPS location. While moving, the ultrasonic sensor continuously scans the area in front of the robot to detect obstacles; whenever an object is too close, the robot stops, turns right using its motors, and then continues forward.

When the robot is near an obstacle, the LED turns on to indicate that something is ahead, and the robot turns right to avoid it. At the measurement point, a servo motor lowers the soil-moisture probe into the ground, allowing the soil sensor to collect an accurate reading before the servo lifts it back up again. The microcontroller stores this value and prepares it for communication with nearby robots or for later analysis. The entire system is powered by a 9V battery, which provides energy to all sensors and actuators so the robot can continue searching for the next measurement point.

Sensors:

. ultrasonic sensor

. IR sensor
. Soil moisture sensor

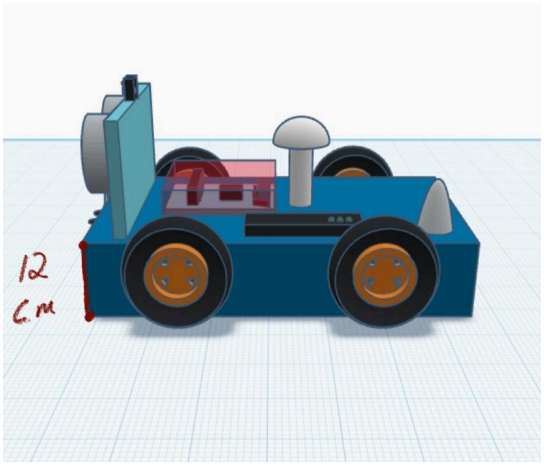
Actuators:

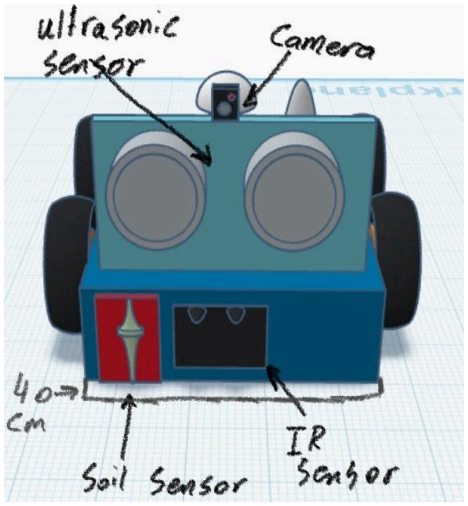
. 2 DC motors (Right and left motor)

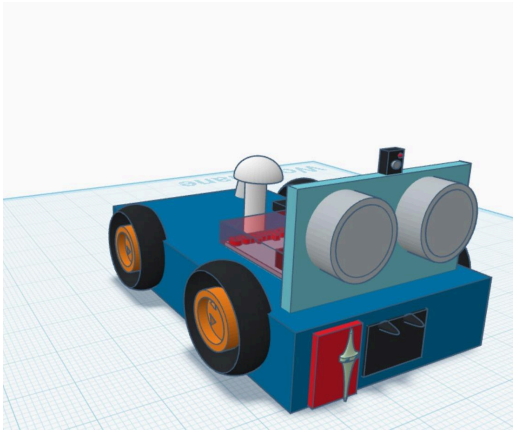
. one Servo motor

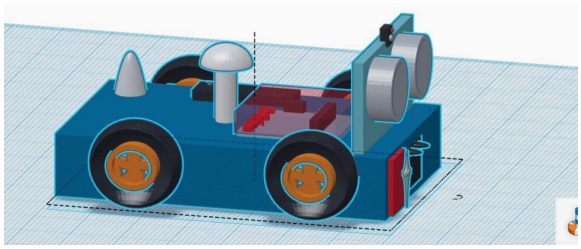
. one LED

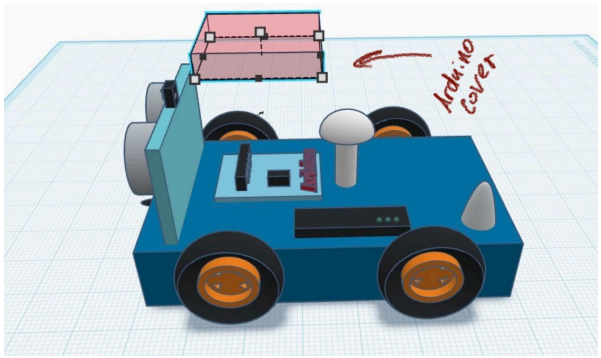
TinkerCad model Figure 5.1.1

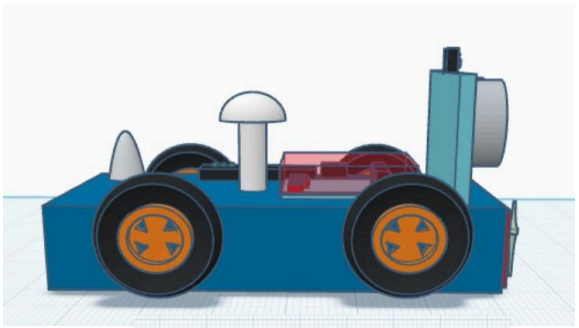


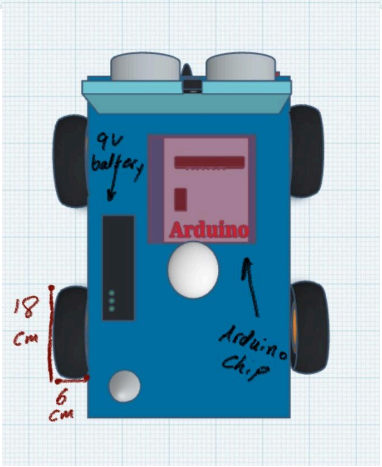


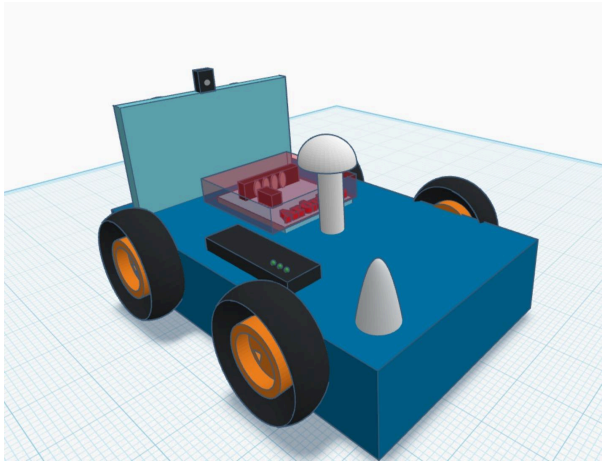












5.1.2 Algorithm

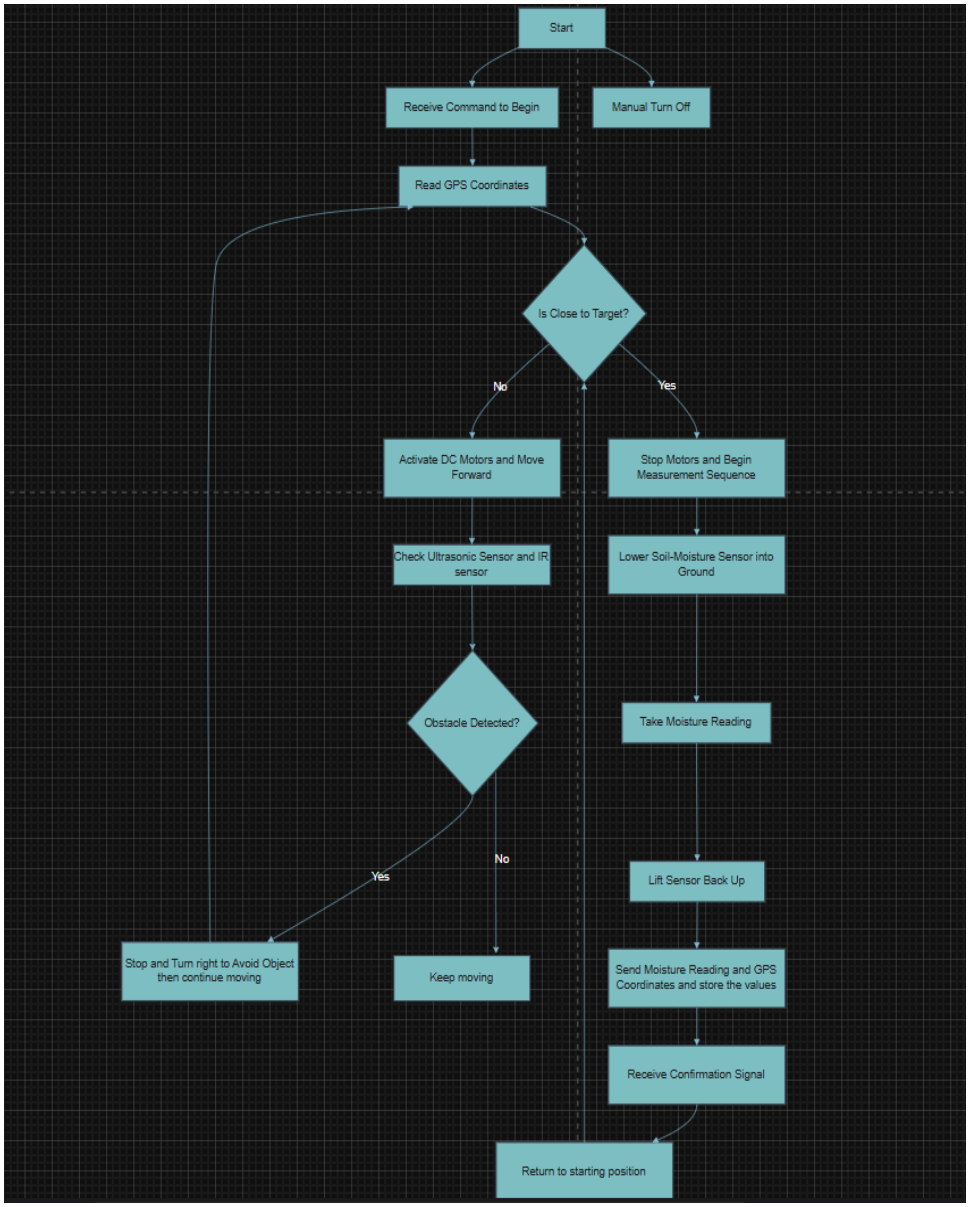
When activated, the robot is turned on then reads the GPS to find its position. If it is not close to the target coordinate, it drives forward using its DC motors while the ultrasonic sensor - IR sensor checks for obstacles. If an object is detected, the robot stops, turns right to avoid it, and continues moving toward the target.

Once the robot reaches the assigned GPS point, it stops and begins the soil-measurement sequence. The servo lowers the moisture sensor into the ground, waits for a stable reading, then raises it back up. The robot stores the data and stores both the moisture value and GPS location to nearby robots or even for research and data. After receiving a simple confirmation signal from the operator, the robot resumes roaming to search for the next measurement point.

Key steps:
Robot being turned on → Move toward target → Avoid obstacles → Measure soil → Store data → Continue to next point.

Figure 5.1.2

Flowchart:



5.2 Patrol Ranger by Janisha Johnson

5.2.1 Description

The Patrol Ranger Robot is built to help monitor forest trails by detecting obstacles, uneven terrain, and other hazards that could affect trail safety. Its main goal is to improve awareness in forest areas by using sensors to spot and respond to changes in the environment. The robot infrared sensors to detect nearby objects and uneven surfaces, while an ultrasonic sensor measures distance to prevent collisions. A visual alert LED light turns on to warn forest rangers or other robots when obstacles are detected.

The patrol robot's body has a rectangular shape and sits slightly elevated from the ground to handle rough terrain. It runs on four large wheels powered by two DC motors, giving its forward and backward motion. A servo motor helps control steering for better balance on slopes. The main control system is an Arduino Uno, which processes all the sensors and adjusts the motors when needed.

Sensors:

- Infrared(IR) Sensor: It detects nearby obstacles or terrain changes within a 20-30 cm range.
- Ultrasonic Sensor: Measures distance to nearby objects and avoids collisions.
- Tilt Sensor: Detects changes in slope greater than 20* This helps for staying balanced on uneven terrain.

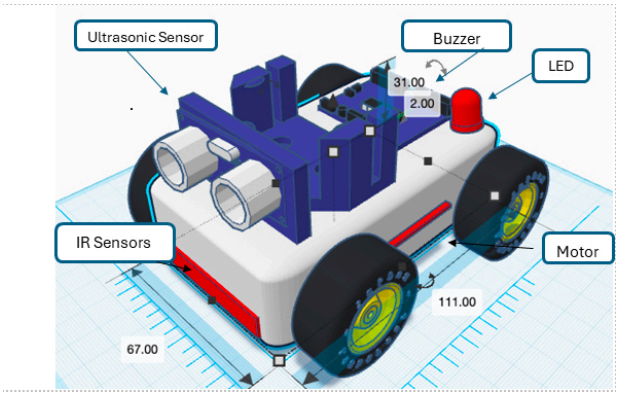
Actuators:

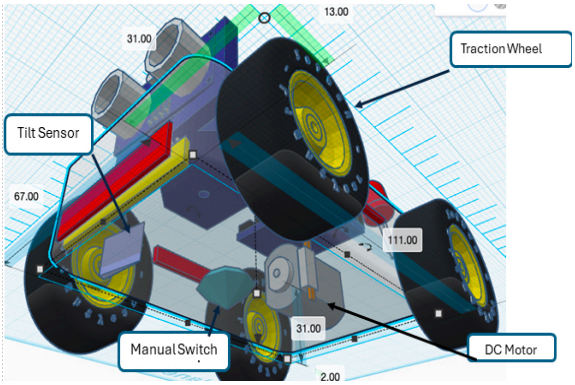
- Two DC Motors: Drives the rear wheels to move the robot forward or backward.
- Servo Motor: adjust steering angle for better control.
- LED Display: Activates when a hazard or obstacle is detected as a visual warning.

Microcontroller:

- Arduino Uno: Processes input from all sensors and sends signals to control the motors and LED Alert.

Figure 5.2.1


Ultrasonic Sensor
Buzzer
LED
31.00
2.00
IR Sensors
Motor
111.00
67.00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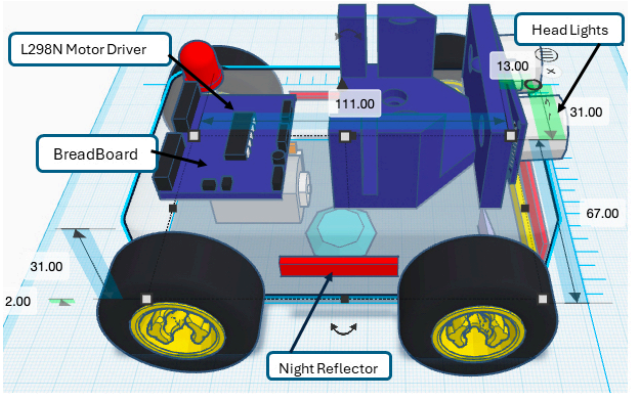
31.00
Traction Wheel
Tilt Sensor
67.00
111.00
31.00
Manual Switch
DC Motor
2.00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Head Lights
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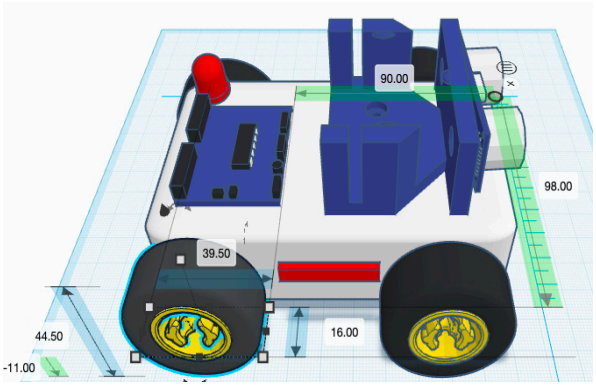
111.00
31.00
BreadBoard
67.00
31.00
2.00
Night Reflector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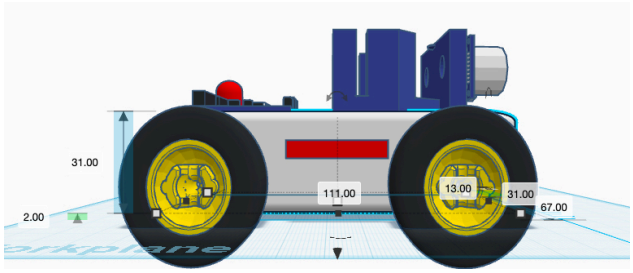
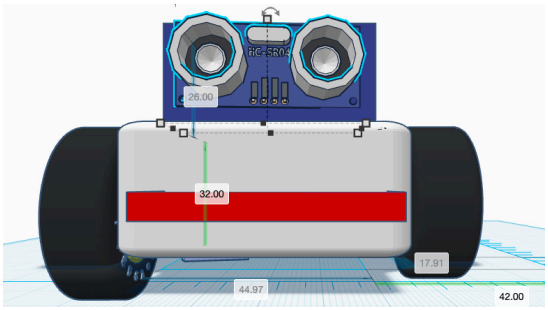
98.00
39.50
44.50
16.00
-11.00


31.00
111.00
13.00
31.00
67.00
2.00


26.00
32.00
17.91
44.97
42.00

5.2.2 Algorithm

The algorithm for the Patrol Ranger Robot explains how the robot reacts to all the different sensor inputs and commands shown in the flowchart. Once the robot is turned on, the Arduino keeps checking whether the ranger is controlling the robot manually or if it should keep navigation on its own. During this whole time, the IR sensor, Ultrasonic sensor, and tilt sensor work together to help the robot stay safe and avoid any dangerous situations on the trail.

Manual Control Mode:

If the ranger presses the on button on the remote, the robot switches into manual mode right away. The Arduino checks which button was pressed and then moves the robot in that direction, either forward, backward, turning right or left, or even stopping. If the button is released, the robot pauses and waits for the next command. But if any other sensors detect anything unsafe, like an obstacle too close or the slope being more than 20*. The robot temporarily turns on the LED that alerts the ranger and the obstacle robot. This lets them know when and where the issue is so it can be addressed, documented and cleared of obstructions.

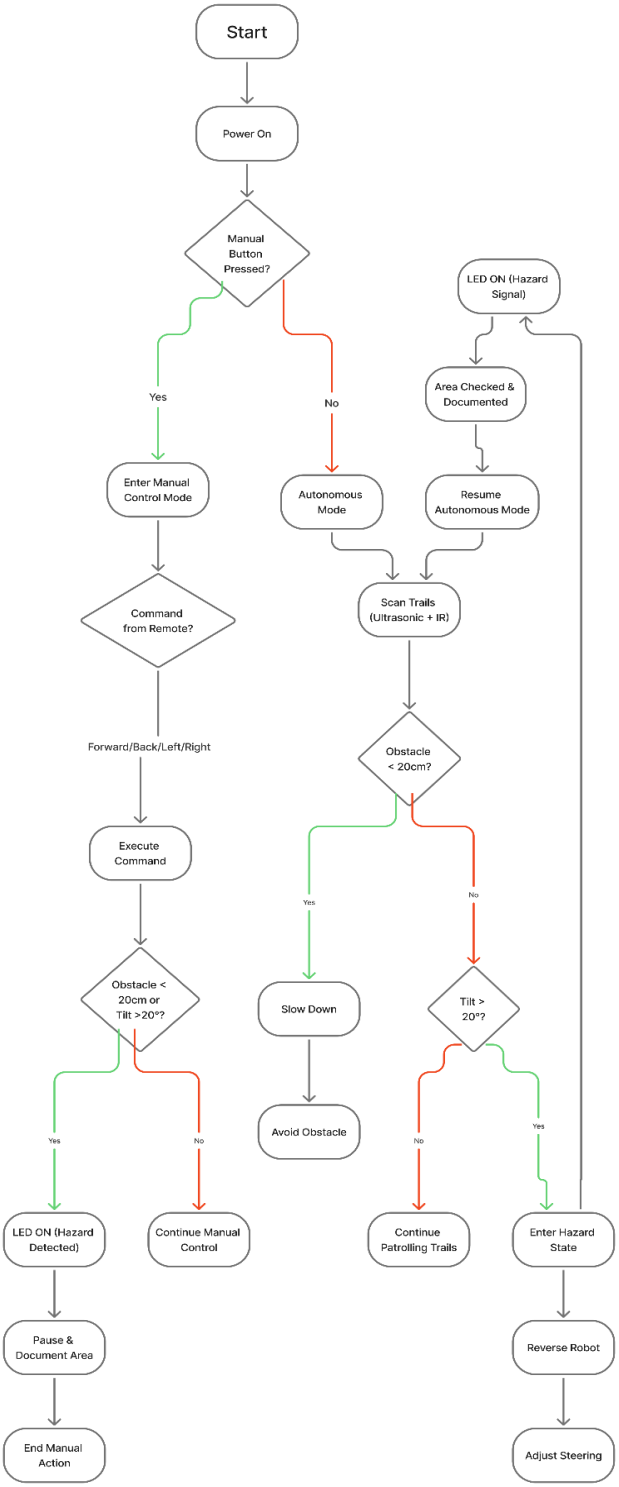
Autonomous Mode:

When no manual commands are being sent, the robot follows its programmed routes. It starts by checking the trail paths for any obstacles. If the ultrasonic sensor detects something closer than 20 cm, the robot slows down and avoids hitting the obstruction. The IR sensor helps it decide whether it needs to turn or reverse. If the tilt sensor shows the slopes is greater than 20* the robot stops and enters the hazards state so it does not tip over. In this mode, it reverses and adjusts its steering until it is stable again. Then it continues to patrol the trails.

Information sent to Ranger and Other Bots:

When the robot enters a hazard state, whether it is from an obstacle, uneven terrain, or a steep slope, the LED alert turns on. This alert is there to let the rangers know the exact area needed to be documented and check for clearance, rangers can evaluate and fix if needed. Once the area is documented the light turns off and the patrol bot continues navigating.

Figure 5.2.2 Flow Chart



5.3 Obstacle Guardian by Mona Breslau

5.3.1 Description

The Obstacle Guardian will be a Sumo-style robot that uses a plow-like attachment to clear obstacles from a forest trail. The design features all-terrain tires, an LED indicator light, and an electronics compartment to house the battery, Arduino unit, and wiring harness. Once turned on, the Obstacle Guardian will remain idling until an IR sensor receives a signal from the Sorting Bot indicating that an obstacle was encountered on the trail. The Obstacle Guardian will then illuminate an LED light and initiate the routing sequence to the obstacle. The Obstacle Guardian will travel along a predefined forest path until the obstacle is encountered. The Robot uses differential rear-wheel drive, with separately controlled Hobby Gearmotors for the rear-left and rear-right wheels to enable turning. An Ultrasonic Distance Sensor will be mounted on the front of the robot and housed behind an observation window in the plow. The distance sensor is calibrated so that when the measured distance from an obstacle is 30 centimeters, the robot initiates the obstacle-clearing sequence. The obstacle-clearing sequence includes a forward maneuver to capture the obstacle in the plow, a 90-degree turn, and another forward maneuver to push the obstacle off of the trail. Once the obstacle-clearance sequence is complete, the Obstacle Guardian initiates a return home sequence. A list of the sensors and actuators are listed below:

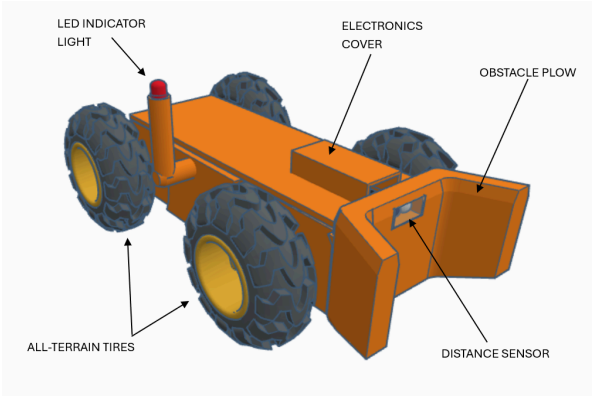
Sensors:
Ultrasonic Distance Sensor
IR Receiver

Actuators:
Left Motor
Right Motor
LED Light

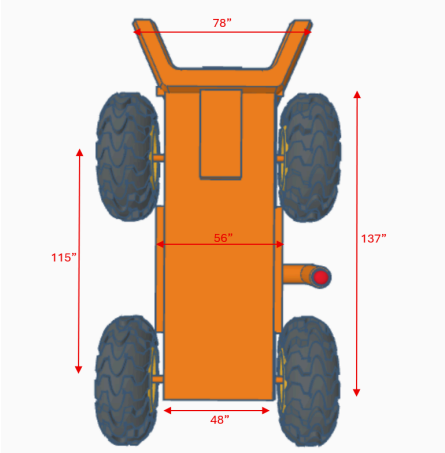
Microcontroller
Arduino UNO 3

Figure 5.3.1.

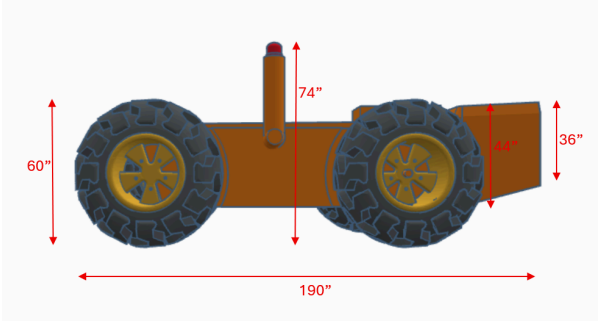
Isometric View



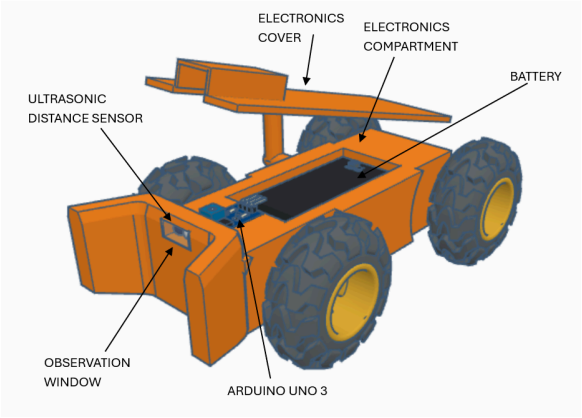
Top View



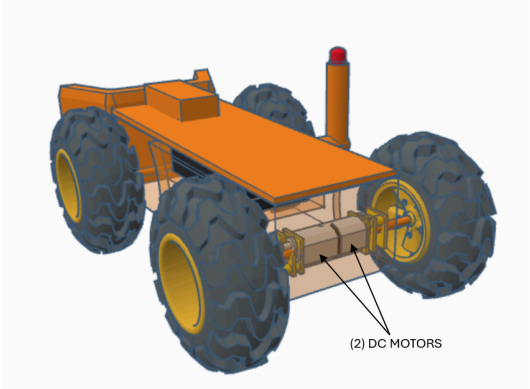
Side View



View With Cover Removed



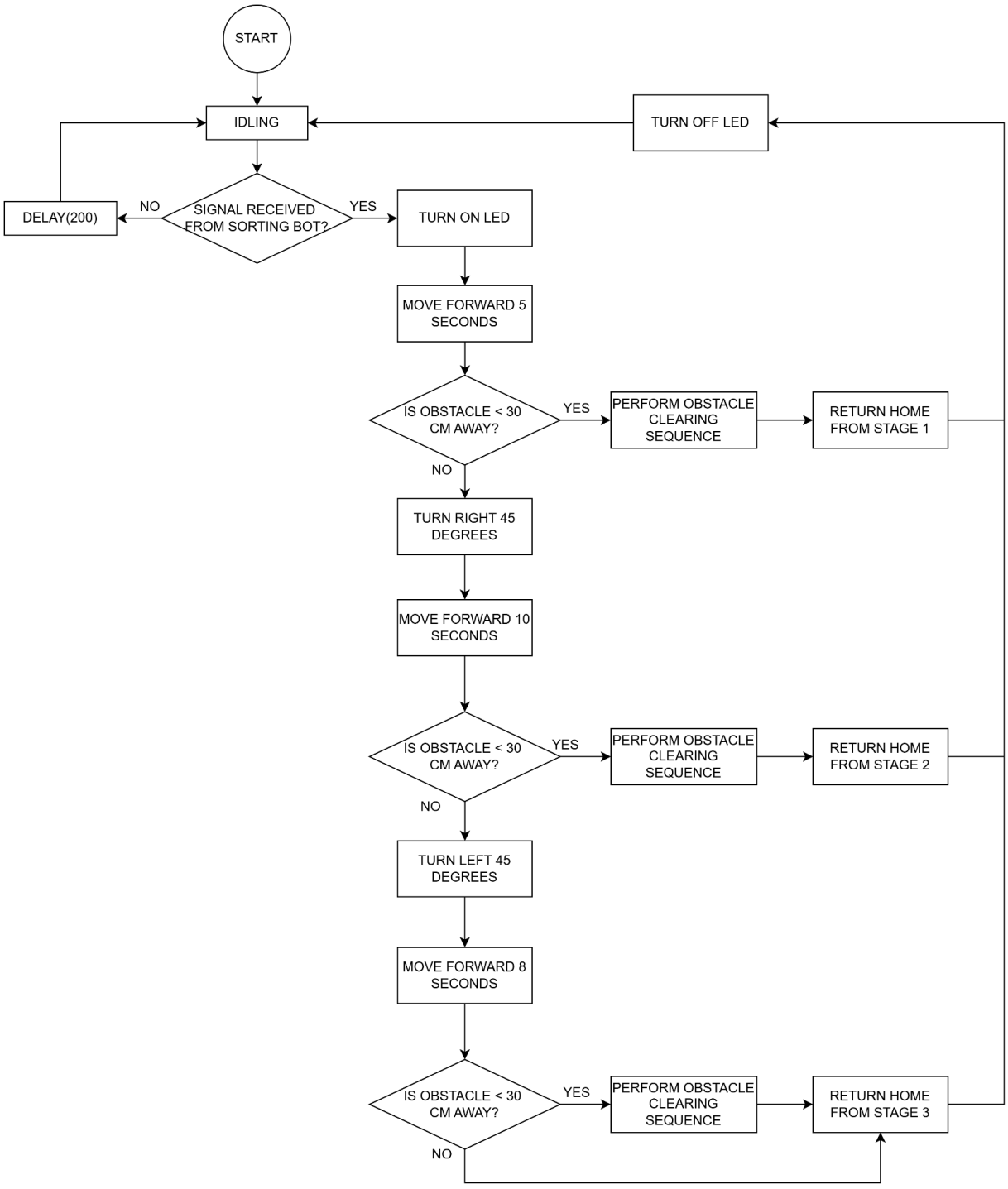
Rear View Showing Motors



5.3.2 Algorithm

When the Obstacle Guardian robot is turned on, it remains in idling mode until it receives an appropriate IR signal from the Patrol Guardian robot. In this circuit, the IR signal is simulated with the “On” button of an IR remote controller. Once that signal is received, indicating that there is an obstacle on the trail, it turns on the LED indicator light and initiates a routing sequence to the obstacle. To simulate a trail path, the robot will move forward for 5 seconds, make a 45-degree right turn, move forward for 10 seconds, make a 45-degree left turn, then move forward another 8 seconds. When the moveForwardWithCheck() function is called, the code performs an if statement contained within a while loop. The if statement is true when the Ultrasonic Distance Sensor detects an object within 30 centimeters, in which case the robot performs an obstacle clearing sequence. This obstacle clearing sequence consists of a forward maneuver to capture the obstacle in the plow, a 90-degree right turn, and another forward movement to push the obstacle off of the trail. Once the Obstacle Guardian pushes the object off the trail, it moves backwards, and then makes another 90-degree right turn so that it is pointed back towards the trail head. After the Obstacle Guardian performs this obstacle clearing sequence at any stage of the route, it initiates a return home sequence by calling the returnHomeRoute() function. This function contains a series of switch cases without breaks, with each switch case representing a return home step. If Obstacle Guardian needs to return from the end of the route, it performs every switch case. If the Obstacle Guardian only needs to return home from the first stage of the route, it only performs the final switch case. Once the Obstacle Guardian returns home, the LED indicator light turns off, and it returns to idle mode in the main function. If the Obstacle Guardian receives another signal from the Sorting Bot, the routing sequence will be re-initiated.

Figure 5.3.2.



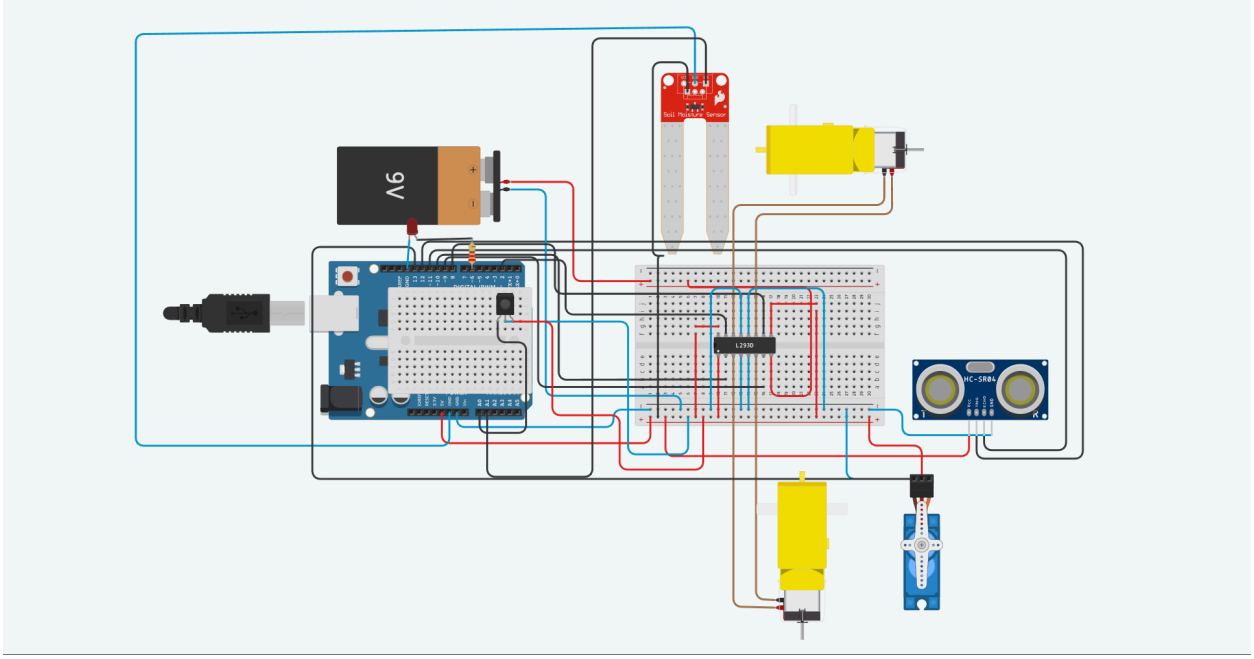
5.2.3 Tinker CAD Simulation *(Omar)*

A TinkerCAD circuit was made for this robot to simulate the algorithm described in the previous subsection. The circuit's snapshot is shown in the Figure 5.2.3 below and the TinkerCAD link for that circuit can be accessed at

TinkerCAD simulation:
https://www.tinkercad.com/things/7xqIRLlQd0l-tremendous-jaiks/editel?returnTo=https%3A%2F%2Fwww.tinkercad.com%2Fdashboard&sharecode=QLcfc-mdB1K61wSGbfJN6rviGOzUAcJt5V_ZHHN5r2w

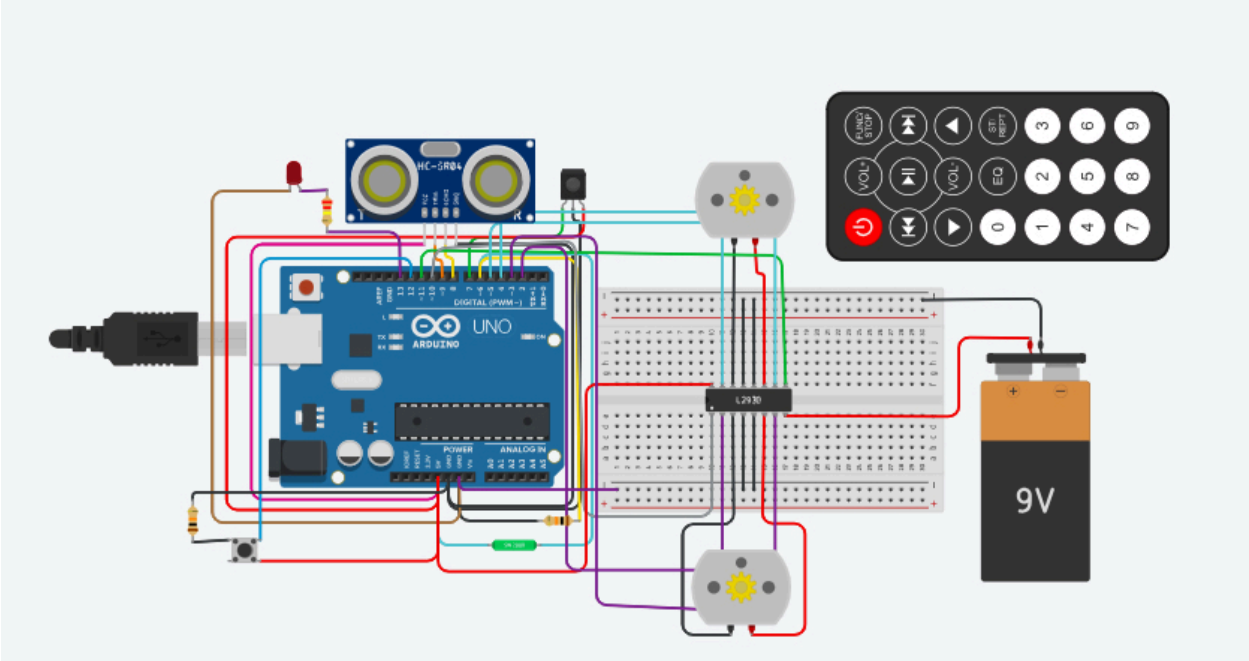
Youtube link:
https://youtu.be/agQ80y1PLqM?si=Uu2f0Ilgp9YDRzjn

Figure 5.2.3.



5.2.3 Tinker CAD Simulation (Janisha)
A TinkerCAD circuit was made for this robot to simulate the algorithm described in the previous subsection. The circuit's snapshot is shown in the Figure 5.2.3 below and the TinkerCAD link for that circuit can be accessed at
https://www.tinkercad.com/things/9QfqPfyXVWO-patrol-ranger?sharecode=rsJ7qT6-AxFTeiMYPhmorXhNXcP0fx7p2BDZzR7Z9II

Also, a short video demonstrating the working circuit and the algorithm can be viewed at this link: https://youtu.be/kL8yjw2-_YY



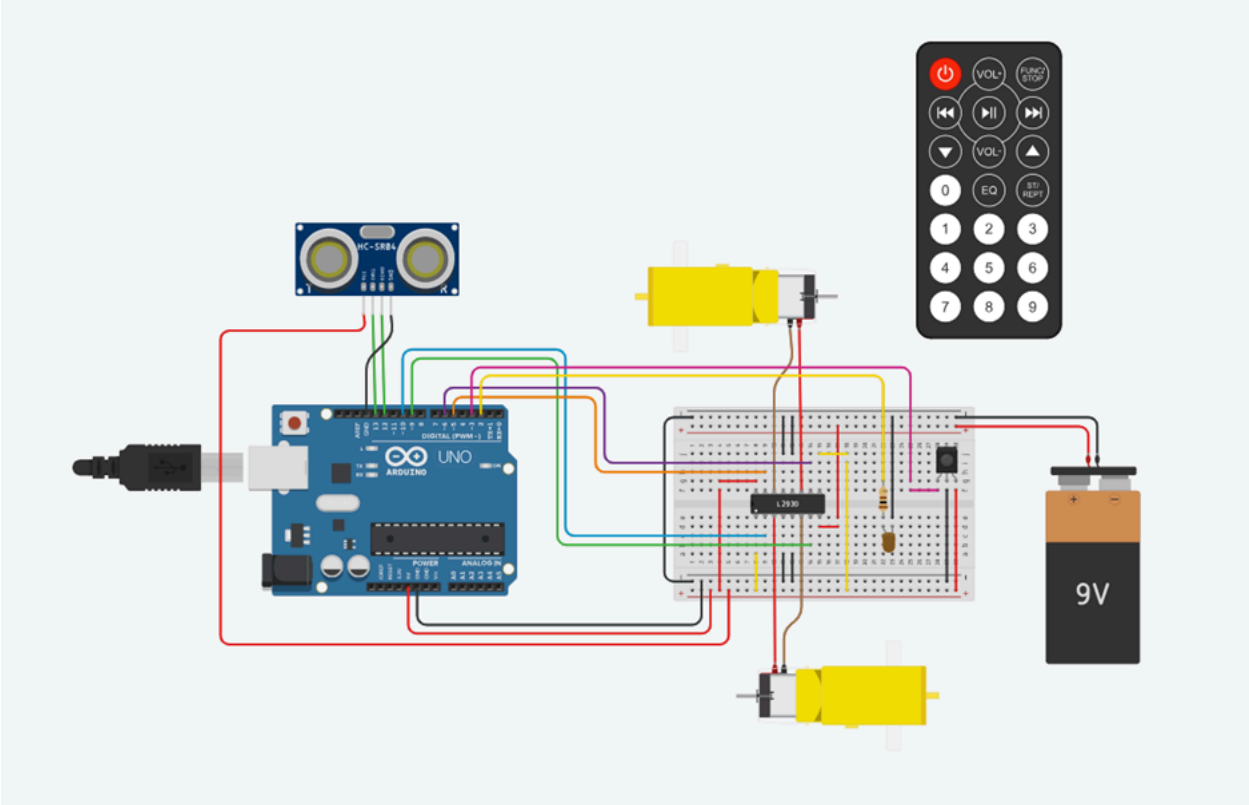
5.3.3 TinkerCAD Simulation *(Mona)*

A TinkerCAD circuit was made for this robot to simulate the algorithm described in the previous subsection. The circuit's snapshot is shown in Figure 5.3.3. below and the TinkerCAD link for that circuit can be accessed at:

https://www.tinkercad.com/things/3odwaB4gtQE-glorious-rottis?sharecode=DZFfO2BCO-3evJe7WMP2n3hFoy4oh5gSXr0JStCqpPM

A short video demonstrating the working circuit and the algorithm can be viewed at this link: https://youtu.be/AHGuAYopMD4

Figure 5.3.3.



6. Robot Collaboration (Janisha)

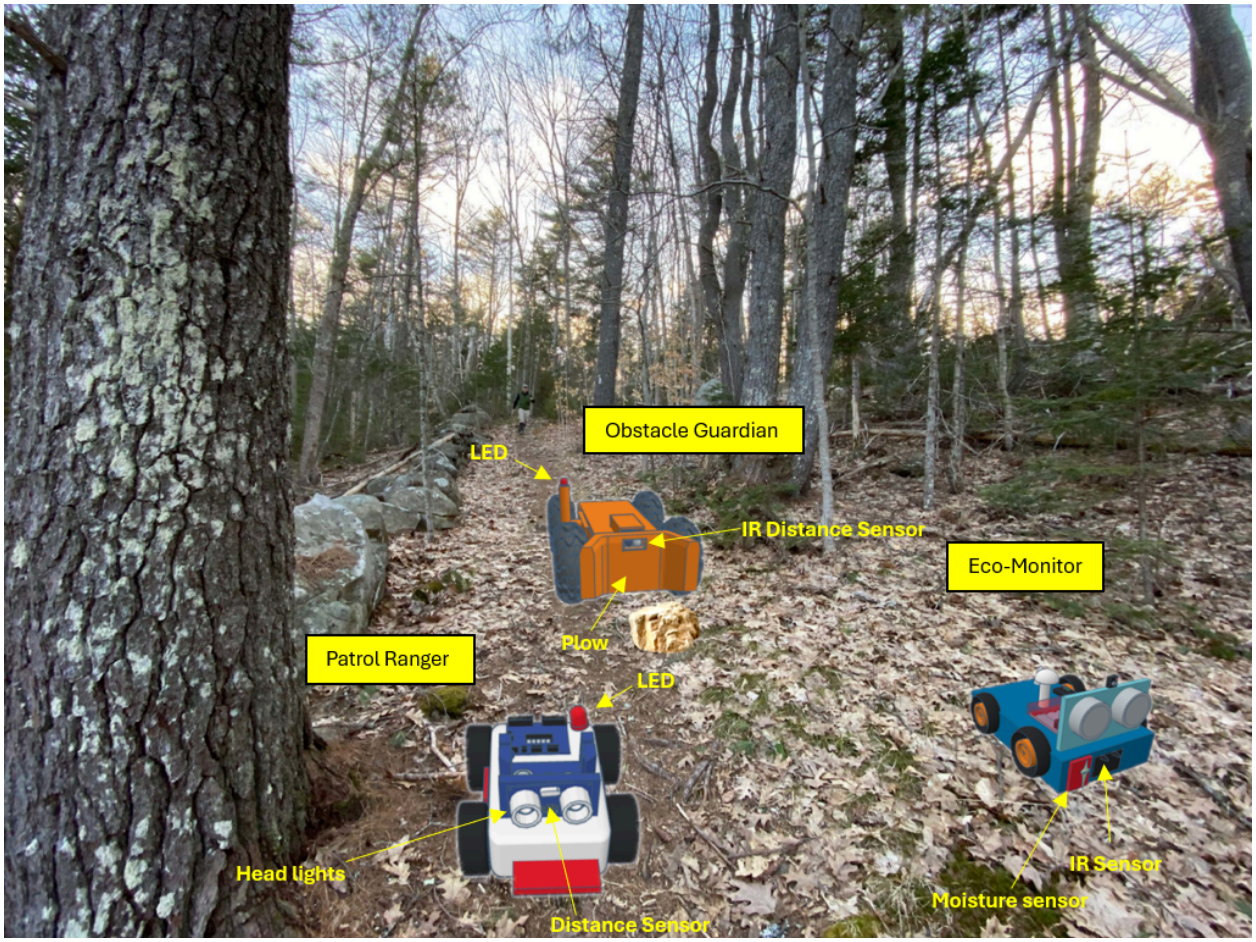
The three robots **Eco-Monitor**, **Patrol Ranger**, and **Obstacle Guardian** work together to keep forest trails safe by sharing information, reacting to each other's signals, and performing different tasks in a coordinated sequence. Below is the step-by-step algorithm describing how they collaborate.

This section describes the sequence of the Teams Operations

1. **The operator turns on the Patrol Ranger.**
    - The Patrol Ranger begins scanning the trail using its ultrasonic, IR, and tilt sensors.
2. **Patrol Ranger moves along the trail and checks for hazards.**
    - If no danger is detected, it continues normal patrol.
    - If an obstacle, uneven terrain, or steep slope (>20°) is detected, it enters a hazard state.
3. **Patrol Ranger sends an IR alert signal.**
    - This IR signal notifies the **Obstacle Guardian** that something on the trail needs clearing.
    - The LED also flashes so nearby rangers know the exact hazard location.
4. **Obstacle Guardian powers on after receiving the signal.**
    - It leaves its idle mode and begins following the predefined trail path toward the hazard.
    - The Obstacle Guardian uses its ultrasonic sensor to verify the obstacle.
5. **Obstacle Guardian reaches the hazard and clears it.**
    - It performs the plow sequence:
        - Moves forward to scoop the obstacle
        - Turns 90°

        - Pushes obstacle off the path
        - Returns to the trail route

6. **Obstacle Guardian sends a “clear” status & turns off its LED.**
    - This indicates that the path is safe again.
7. **Eco-Monitor begins its task once the path is confirmed clear.**
    - The Eco-Monitor reads its GPS position
    - Travels toward its measurement point
    - Uses ultrasonic & IR to avoid new obstacles
8. **Eco-Monitor performs soil-moisture measurement.**
    - Lowers probe
    - Collects data
    - Stores the information
    - Sends its measurement data for later analysis or communication
9. **Eco-Monitor moves to the next GPS point and repeats.**
10. **Patrol Ranger resumes patrol behind the other robots.**
    - Ensures the trail remains safe after the Obstacle Guardian clears obstacles
    - Reports new hazards if they appear again



7. Conclusion

The three-robot system we designed adds strong value to forest management by meeting all major system and robot-specific requirements described in Section 2.3. The Eco-Monitor collects soil-moisture data at accurate GPS locations, the Patrol Ranger identifies unsafe trail conditions, and the Obstacle Guardian removes objects that block forest paths. Together, they operate autonomously over a 100 m × 100 m area, communicate within a 100 m radius, work safely on uneven terrain, and stay visible to people in both day and night conditions. These combined abilities directly help address the maintenance problems outlined in Section 1, such as trail hazards, debris buildup, pollution, and the slow and labor-intensive nature of current forest-maintenance practices.

Compared with current tools and solutions discussed in Section 1 such as manual litter pickers, handheld trash bag holders, simple collapsible collection bins, and traditional hand tools, our system provides a much more complete and automated approach. While existing solutions depend fully on human labor and require workers to walk long trails to remove debris or inspect soil, our robots can perform these tasks continuously and with less physical effort. Because forest maintenance backlogs cost hundreds of millions of dollars each year, even a modest reduction in labor hours can have meaningful economic benefits for agencies. Our

robots are built from affordable components such as ultrasonic sensors, IR sensors, DC motors, and Arduino controllers, which makes the system low-cost compared to high-end commercial robotic solutions. These advantages make our design practical, scalable, and a more efficient option for agencies who want to reduce maintenance delays and improve forest safety.

This report covered the full development of the robotic system. Section 1 was the introduction and it described the challenges in forest trail maintenance and the limitations of existing manual methods. Section 2 listed the requirements for the overall system and each robot. Section 3 outlined the design process, and Section 4 was narrowing our choices and keeping the best ideas we have. Section 5 was proposing robot designs of the Eco-Monitor, Patrol Ranger, and Obstacle Guardian , while Section 6 presented was basically how the robots work together and accomplish the goal already set . The main problem addressed was the difficulty and cost of maintaining large forest areas. We proposed a coordinated three-robot solution to monitor soil conditions, detect hazards, and clear obstacles automatically. A video summarizing this project as a pitch to customers is included in Appendix B *after we record the video we will add it to the appendix*.

Throughout this project, we learned many important lessons. We discovered the importance of early prototyping and frequent testing, especially when working with sensors on uneven terrain. Communication between robots also proved challenging and taught us to simplify our approach and plan for backup methods. If given another chance, we would improve battery capacity, increase the strength of the Obstacle Guardian’s clearing mechanism, and expand testing soil moisture to more environments. These experiences helped us understand how to design more reliable robotic systems and how to improve teamwork and problem-solving during technical projects.

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Appendix A: Gantt Chart (Group)

https://docs.google.com/spreadsheets/d/1c4c0E8iFUkD9Z5kMtyY5WkPM4Lv8kPyihZ0T7NTHcAk/edit?usp=sharing

Appendix B: Pitch Video

Here is a video summarizing this project as a pitch to the customers:
https://youtu.be/xqhZXyMk4mA?si=EE0mxfLNtXyCGusi